Course Title: Ethical Hacking

Objective

The objective of this experiment is to perform information gathering using Nmap, a network scanning tool used for host and service discovery, network mapping, port scanning, operating system detection, and service enumeration. Students will use Nmap against an authorised target (DVWA) within the isolated lab network to identify live hosts, open ports, running services, and the underlying operating system.

Methodology

1. Introduce host and service discovery scanning and its role in the vulnerability assessment lifecycle.
2. Introduce network mapping and enumeration concepts.
3. Install and verify Nmap on Kali Linux.
4. Perform host discovery to identify live systems on the lab network.
5. Perform port scanning to identify open ports on the DVWA target.
6. Perform OS detection and service/version enumeration on discovered open ports.
7. Analyze and document the scan results.

Expected Outcome

By completing this experiment, you will be able to:

1. Explain the role of host and service discovery in a security assessment.
2. Install and verify Nmap on Kali Linux.
3. Perform basic and advanced Nmap scans using common flags.
4. Detect the operating system and enumerate running services on a target.
5. Analyze and interpret Nmap scan results to identify potential attack surfaces.

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1. Introduction to Host and Service Discovery Scanning

Host and service discovery scanning is the process of identifying which systems on a network are active ("alive") and which network services they are running. It is the first active scanning step in a security assessment, following the passive reconnaissance and footprinting covered in earlier weeks.

1.1. Uses of Nmap

1. Confirms which hosts actually exist before further scanning is attempted.
2. Identifies open ports, which represent potential entry points into a system.
3. Forms the foundation for later vulnerability assessment and exploitation phases.

2. Introduction to Network Mapping and Enumeration

Network mapping is the process of building a visual or logical representation of a network's structure — its hosts, open ports, and services. Enumeration goes a step further, actively extracting detailed information such as service names, version numbers, and, where possible, the operating system in use.

2.1. Important Note

Unlike the passive OSINT tools used in Weeks 3–5, Nmap is an active scanning tool. It sends packets directly to the target and is detectable by the target's own logs, firewalls, and intrusion detection systems. Only scan targets you own or are explicitly authorised to test — in this course, that means DVWA or another designated training target only.

3. Introduction to Nmap

Nmap (Network Mapper) is a free, open-source tool used for network discovery and security auditing. It is one of the most widely used tools in the industry for host discovery, port scanning, service/version detection, and OS fingerprinting.

3.1. What Nmap Can Do

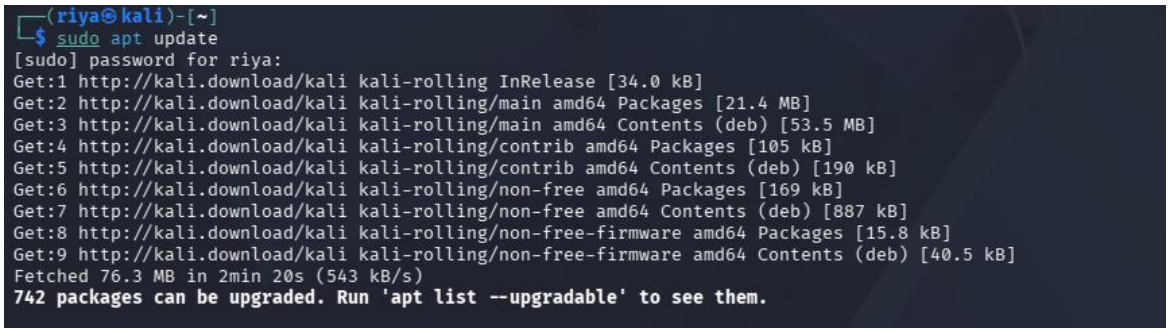
1. Discover live hosts on a network.
2. Identify open, closed, and filtered ports.
3. Detect the service and version running on each open port.
4. Attempt to fingerprint the target's operating system.

4. Installing Nmap

Step 1: Update the Package List

```
sudo apt update
```

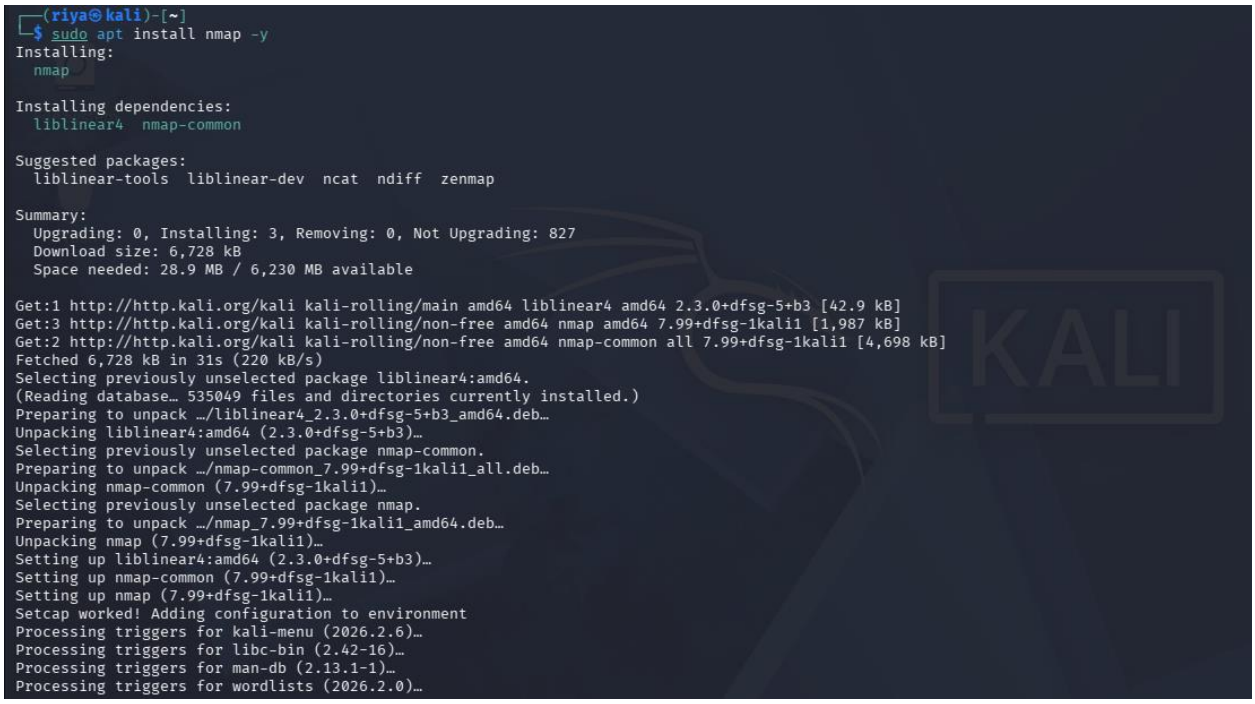
Updates the package repository information.



Step 2: Install Nmap

```
sudo apt install nmap -y
```

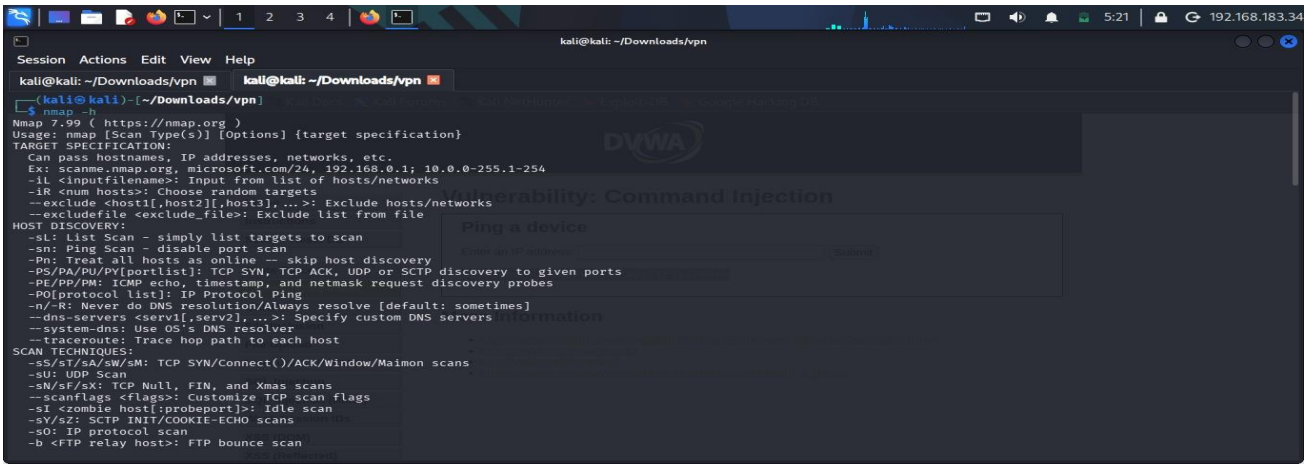
Installs Nmap (usually already present by default on Kali Linux).



Step 3: Verify Installation

```
nmap --h
```

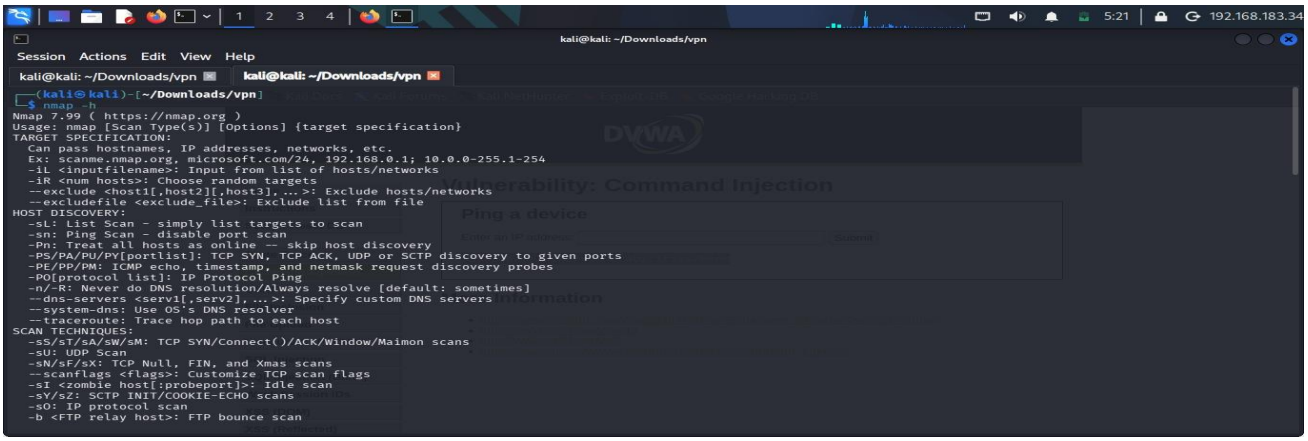
Displays the installed Nmap version.

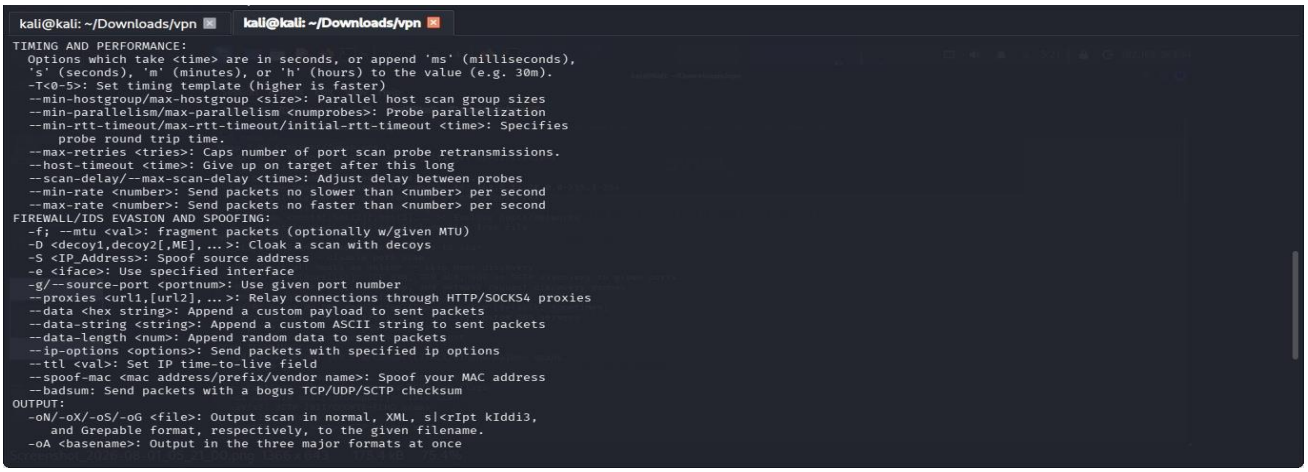


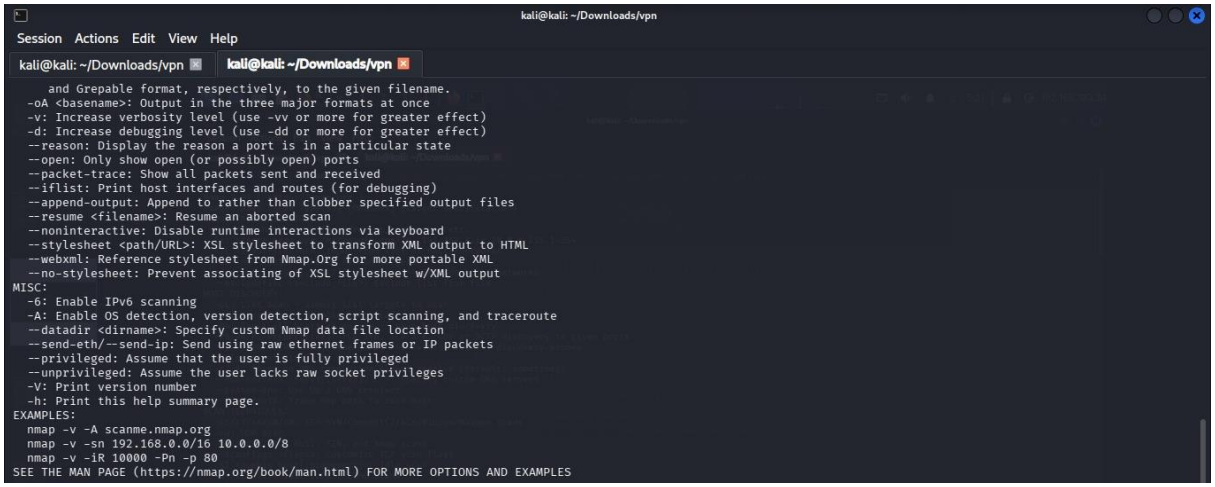
5. Nmap Command Syntax

Basic syntax:

```
nmap [scan type] [options] <target>
```







Example: nmap -sS -sV -O 10.49.154.234 combines a SYN scan, version detection, and OS detection in a single command.



6. Host Discovery

Step 1: Confirm the Lab Network Range

Identify your isolated Host-only network range (e.g. 192.168.56.0/24) using ip a on the Kali VM.

```
┌──(riya㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ab:06:ed brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 86309sec preferred_lft 86309sec
    inet6 fd17:625c:f037:2:53a5:2c4b:d067:ca9a/64 scope global temporary dynamic
       valid_lft 86311sec preferred_lft 14311sec
    inet6 fd17:625c:f037:2:a00:27ff:feab:6ed/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86311sec preferred_lft 14311sec
    inet6 fe80::a00:27ff:feab:6ed/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Step 2: Perform a Ping Scan

```
nmap -sn 10.0.2.0/24
```

-sn performs host discovery only (no port scan), listing which hosts on the range are currently alive.

```
┌──(riya㉿kali)-[~]
└─$ nmap -sn 10.0.2.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 02:39 -0400
Nmap scan report for 10.0.2.2
Host is up (0.00043s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 10.0.2.3
Host is up (0.00040s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 10.0.2.15
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 7.20 seconds
```

7. Port Scanning

Step 1: Perform a Basic Port Scan

```
nmap 8.8.8.8
```

Scans the target's most common 1000 ports and reports which are open, closed, or filtered.

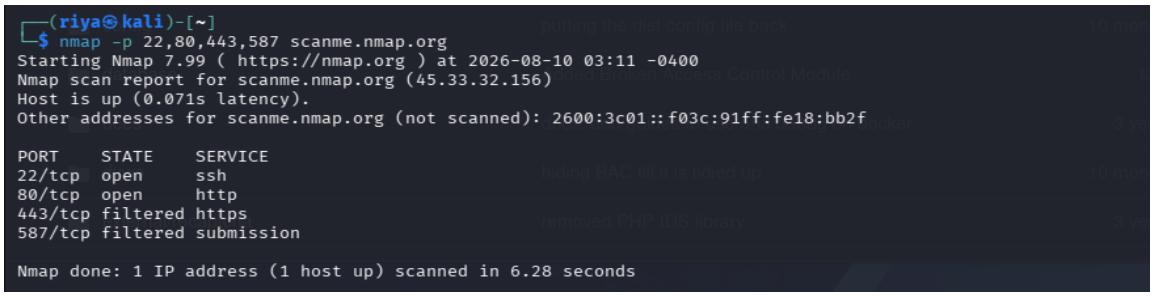
```
┌──(riya㉿kali)-[~]
└─$ nmap 8.8.8.8
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 02:52 -0400
Nmap scan report for dns.google (8.8.8.8)
Host is up (0.014s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
53/tcp  open  domain
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 7.61 seconds
```

Step 2: Perform a Full Port Scan

```
nmap -p- 22,80,443,587
```

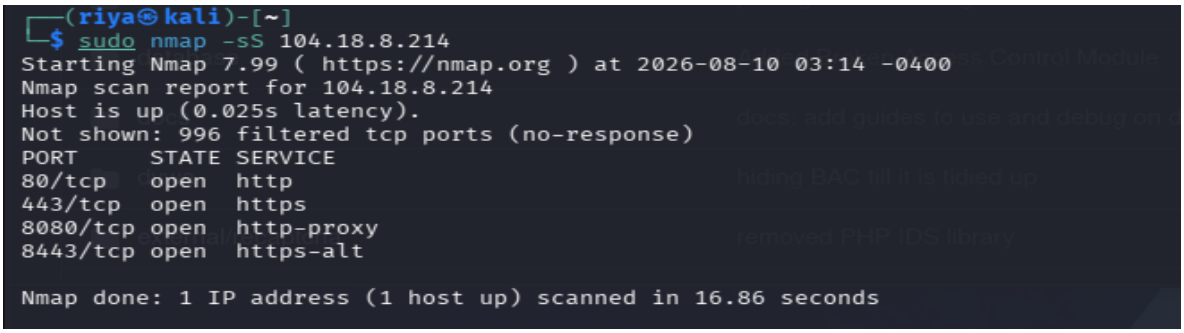
-p- scans all 65535 ports rather than only the default top 1000, at the cost of a longer scan time.



Step 3: Perform a SYN (Stealth) Scan

```
sudo nmap -sS 192.168.56.10
```

-sS performs a SYN scan, which is faster and less easily logged than a full TCP connect scan. Requires root privileges.



8. OS Detection

Step 1: Run OS Detection

```
sudo nmap -O 10.49.154.234
```

-O attempts to fingerprint the target's operating system based on how it responds to specially crafted packets.

```
┌──(kali㉿kali)-[~/Downloads/vpn]
└─$ nmap -O 10.49.154.234
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-01 05:45 -0400
Nmap scan report for 10.49.154.234
Host is up (0.039s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
80/tcp   open  http
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
8009/tcp open  ajp13
8080/tcp open  http-proxy
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.99%E=4%D=8/1%OT=22%CT=1%CU=43792%PV=Y%DS=3%DC=I%G=Y%TM=6A6DC05E
OS:%P=x86_64-pc-linux-gnu)SEQ(SP=102%GCD=1%ISR=10E%TI=Z%CI=Z%II=I%TS=A)SEQ(
OS:SP=107%GCD=1%ISR=10B%TI=Z%CI=Z%II=I%TS=A)SEQ(SP=107%GCD=1%ISR=10D%TI=Z%C
OS:I=Z%II=I%TS=A)SEQ(SP=FA%GCD=1%ISR=F6%TI=Z%CI=Z%II=I%TS=A)OPS(O1=M4E8ST11
OS:NW7%O2=M4E8ST11NW7%O3=M4E8NNT11NW7%O4=M4E8ST11NW7%O5=M4E8ST11NW7%O6=M4E8
OS:ST11)WIN(W1=F4B3%W2=F4B3%W3=F4B3%W4=F4B3%W5=F4B3%W6=F4B3)ECN(R=Y%DF=Y%T=
OS:40%W=F507%O=M4E8NNSNW7%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=O%A=S+%F=AS%RD=0%Q=)T2
OS:(R=N)T3(R=N)T4(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40
OS:%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q
OS:=)T7(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164
OS:%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 3 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.85 seconds
```

Step 2: Interpret the Confidence Level

Nmap reports an OS guess along with an accuracy percentage. Lower percentages indicate the fingerprint was not a strong match and should be treated as indicative rather than certain.

9. Service and Version Enumeration

Step 1: Run Version Detection

```
nmap -sV 10.49.154.234
```

-sV probes each open port to determine the specific service and version number running on it.

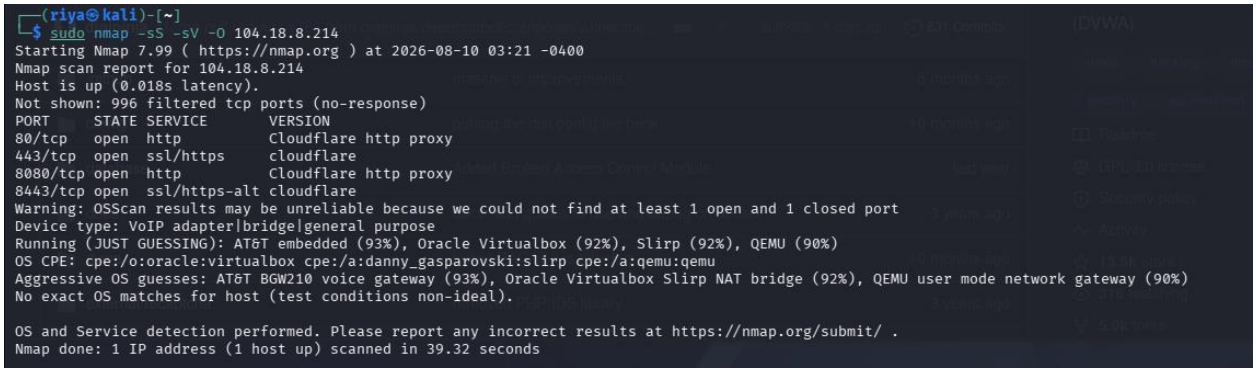
```
┌──(kali㉿kali)-[~/Downloads/vpn]
└─$ nmap -sV 10.49.154.234
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-01 05:39 -0400
Nmap scan report for 10.49.154.234
Host is up (0.12s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
22/tcp   open  ssh         OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)
80/tcp   open  http        Apache httpd 2.4.41 ((Ubuntu))
139/tcp  open  netbios-ssn Samba smbd 4
445/tcp  open  netbios-ssn Samba smbd 4
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8080/tcp open  http        Apache Tomcat 9.0.7
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.35 seconds
```

Step 2: Combine Scans for a Full Profile

```
sudo nmap -sS -sV -O 10.49.154.234
```

Combines a SYN scan, service/version detection, and OS detection into a single comprehensive scan.



Step 3: Run an Aggressive Scan

```
sudo nmap -A 10.49.154.234
```

Combines OS detection, version detection, script scanning, and traceroute into a single comprehensive scan using the -A flag.

